

Problem Statement No. 1

Nutrition Agent

The Challenge

Good nutrition is the foundation of health, but individuals often struggle with diet planning, portion control, and tracking nutritional intake. Information about healthy diets, disease-specific nutrition, and food composition is scattered across unreliable sources. Without personalized, real-time guidance, many fail to maintain balanced diets or make informed food choices.

The Objective

Build an AI-powered Nutrition Agent that delivers personalized dietary insights and guidance for individuals, families, and health professionals.

Core Requirements

- 1. Personalized Diet Plans** — Recommend meals based on age, health conditions, allergies, cultural preferences, and fitness goals.
- 2. Food Data RAG Retrieval** — Fetch and summarize nutritional values such as calories, macronutrients, and vitamins from reliable sources.
- 3. Diet Tracking & Feedback** — Allow users to log meals via text, images, or voice, and provide instant nutritional analysis.
- 4. Preventive Health Focus** — Suggest diets for chronic disease management, such as diabetes-friendly or heart-healthy plans.

5. Multi-Agent System

Agent	Responsibility
Nutrition Knowledge Agent	Fetches nutritional data.
Diet Recommendation Agent	Creates personalized meal plans.
Health Advisory Agent	Provides preventive health suggestions.
Food Log & Feedback Agent	Tracks meals and provides nutritional analysis.

6. Visualization Dashboard

Show daily and weekly nutrient intake, deficiencies, and progress toward health goals.

Mandatory Technology Stack

Component	Requirement
AI Development Platform	IBM Bob

AI / Foundation Models	IBM Models
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Goal: Deliver an AI-powered, personalized nutrition assistant that combines reliable food-data retrieval, multi-agent reasoning, meal tracking, preventive health guidance, and clear nutritional visualizations.